Temperature and particle motion

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use a particle model to distinguish average motion from total energy.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 3.*
Electromagnetic radiation can travel in a vacuum; conduction/convection require matter.

02. Worked example: explain the model
Warmer sample has greater average motion, not necessarily every particle faster than every colder one.

03. Read and interpret the model
Hotter sample has larger average kinetic energy; not every particle has the same speed.

04. Apply the physics
Factor 2.

05. Explain your reasoning
Must use kelvin: 293 to 313 K is only a factor 313/293≈1.068.

06. Review: connect an earlier idea *Review: Unit 4, Day 16.*
Same work; the slower student has 1/4 of the faster student's average power.

07. Exit check
No; same average per particle, roughly twice total translational energy for the same ideal gas conditions.

08. STRETCH: test the model
No; temperature relates to average kinetic energy, not volume alone. Mass/species also affects typical speeds.

Temperature and particle motion

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Use a particle model to distinguish average motion from total energy.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 3.*
Net thermal radiation transfers energy to the cooler radiative environment. Absolute zero is not required.

02. Worked example: explain the model
Warmer sample has greater average motion, not necessarily every particle faster than every colder one.

03. Read and interpret the model
Hotter sample has larger average kinetic energy; not every particle has the same speed.

04. Apply the physics
6.21×10^{-21} J; doubles to 1.242×10^{-20} J.

05. Explain your reasoning
Must use kelvin: 293 to 313 K is only a factor $313/293\approx1.068$.

06. Review: connect an earlier idea *Review: Unit 4, Day 16.*
$P=Fv=240$ W; $W=Pt=1920$ J.

07. Exit check
$\sqrt{2}$, since kinetic energy is proportional to speed squared.

08. STRETCH: test the model
Speed doubles; the relationship uses absolute temperature. Celsius zero is not zero thermal motion.